

AGRICULTURAL TECHNOLOGIES COMPETITION UNIVERSITY AND ABOVE LEVEL PRELIMINARY EVALUATION REPORT

**CATEGORY NAME: AGRICULTURE
TECHNOLOGIES COMPETITION**

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1. PROJECT SUMMARY (20 POINTS)

1.1. Project Summary and Project Scope (10 POINTS)

MoonHarvest – Precision Agriculture Monitoring System is a cross-platform Ground Control Station (GCS) that integrates real-time computer vision with UAV technology for crop monitoring. The project addresses the limitations of manual field inspection, which is time-consuming and often leads to late detection of crop stress, pests, and irrigation issues. By processing live UAV video using a YOLO-based model, the system provides instant analysis directly within the GCS.

The system targets smallholder and medium-scale farmers as well as agricultural operators who need efficient and accessible monitoring tools. A UAV transmits video and telemetry via MAVLink, while the GCS analyzes and displays results in real time on Windows, Linux, and Android using a single C# codebase. This enables faster decision-making, reduces crop losses, and supports efficient resource use, aligning with sustainable and technology-driven agriculture.

1.2. Project Purpose and Social Benefit (10 POINTS)

MoonHarvest gives farmers an automated, real-time crop monitoring tool without manual field walking. The UAV surveys the field while the GCS processes the live feed using computer vision, identifying anomalies instantly on any device the farmer already owns laptop or Android phone.

Social benefit operates on three levels: (1) faster detection enables earlier intervention, reducing crop losses; (2) precise localization of affected areas enables targeted pesticide and fertilizer application, reducing chemical overuse and environmental impact; (3) native Android support removes the device barrier for farmers who only have a smartphone. The project uses entirely open-source tools Uno Platform, OpenCVSharp, ONNX Runtime, MAVLink with no licensing costs.

2. PROBLEM DEFINITION (15 POINTS)

Manual field inspection requires 4-6 person-hours per 10 hectares, often detecting stress too late. Globally, late detection causes 40% crop loss (USD 220 billion) and drives 15-25% excessive pesticide use, degrading environments (FAO, 2025). While UAVs offer aerial views, current workflows are fragmented and slow: farmers must fly the drone, extract SD cards, and use expensive, cloud-dependent software for post-flight analysis (Panotra, 2025; Zaivyi, 2025). Standard Ground Control Stations (GCS) only manage flight, lacking integrated crop health analysis. This creates a critical latency bottleneck, leaving smallholders without an affordable, real-time tool to make instant interventions in the field.

3. SOLUTION PROPOSAL AND ORIGINAL THINKING (30 POINTS)

3.1. Solution Idea (12 POINTS)

MoonHarvest consists of a UAV platform and the MoonHarvest GCS. The UAV (ArduPilot/PX4) transmits live video and telemetry via MAVLink over UDP, TCP, or serial. The GCS built with Uno Platform on .NET compiles one C# codebase to

native Windows, Linux, and Android applications, ensuring accessibility on any device.

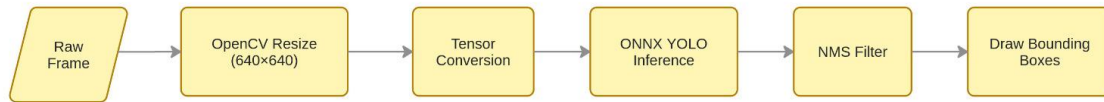


Figure 1. MoonHarvest System Architecture — UAV to Cross-Platform GCS data flow.

The GCS integrates four key modules: (a) YOLO Crop Detection: Each camera frame is processed through a YOLO pipeline (ONNX Runtime) for real-time detection, with bounding boxes and confidence scores displayed on the live feed. Model (.onnx) and classes are configurable at runtime to support different crop types. (b) Vegetation Analysis Overlay: An OpenCV-based module applies color transformations to generate a real-time map of vegetation health. (c) MAVLink Telemetry: Provides real-time data including attitude, GPS, altitude, airspeed, and flight mode via UDP, TCP, or serial connections. (d) Waypoint & Geofence: Enables waypoint-based autonomous flight planning with boundary enforcement, real-time alerts, and telemetry logging.



Figure 2. MoonHarvest GCS Desktop Interface showing real-time YOLO detection overlay and telemetry.

3.2. Comparison with Existing Solutions (8 POINTS)

MoonHarvest integrates UAV control, live video, and automated crop analysis into a single cross-platform application eliminating the workflow gap that makes existing solutions impractical for real-time field decisions.

Feature	Satellite	DJI / Commercial	QGC / Mission Planner	MoonHarvest
Real-time CV analysis	✗	✗ (separate app)	✗	✓ YOLO built-in
Cross-platform Win/Linux/Android	N/A	✗ proprietary	✗ limited	✓ single codebase
Open-source & customizable	✗	✗	Partial	✓
Native Android GCS	✗	✗	✗	✓

3.3. Original Thinking (10 POINTS)

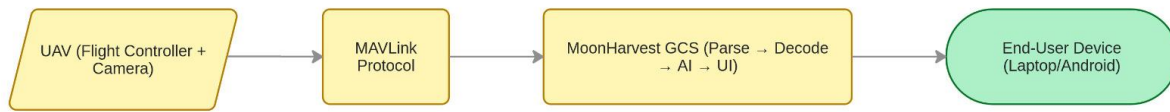


Figure 3. MoonHarvest low-latency Computer Vision pipeline executing per frame

(1) Zero-Internet Edge AI: MoonHarvest runs YOLO inference entirely offline on the local GCS. This eliminates cloud dependency and post-flight analysis, ensuring instant decision-making even in remote fields without internet access. (2) One codebase, three platforms: Using Uno Platform, a single C# codebase runs natively on Windows, Linux, and Android with full hardware access, unlike typical web-based GCS solutions. (3) Future-Proof Adaptability: YOLO models (onnx) and thresholds are dynamically swappable at runtime. This allows users to instantly adapt the same GCS for different seasons, crops, or emerging pests without requiring any software updates or coding skills.

4. PROJECT DEVELOPMENT PROCESS AND WORKING METHOD (15 POINTS)

Stage 1 Concept (Month 1): Literature review identified the lack of real-time analysis in existing UAVs. Stage 2 Design (Month 2): Modular GCS designed via Uno Platform, OpenCV, ONNX, and MAVLink. Stage 3 Development (Month 3-4): Integrated MAVLink, camera pipeline, and YOLO. Risk Mitigation: INT8 quantization prevents Android bottlenecks. Stage 4 Testing (Month 5-6): Conducted SITL simulations and cross-platform tests, targeting >15 FPS on mid-range Androids (detection accuracy), performance testing (frame rate under CV load), and cross-platform compatibility (Windows, Linux, Android). ArduPilot SITL used for UAV simulation.

5. MARKET EVALUATION AND INNOVATION (20 POINTS)

Precision agriculture is a \$9.5B market (12.7% CAGR). Commercial drones (\$10k-\$30k) exclude smallholders, while open-source tools lack automated analysis. MoonHarvest fills this gap as a license-free system on existing Androids/laptops, with compatible UAVs costing just \$300-\$800. Distribution: Initial adoption via agricultural cooperatives.

Technical Feasibility: Built via open-source tools (Uno Platform, OpenCVSharp, ONNX, MAVLink) needing no proprietary hardware. Sustainability: Supports dynamic YOLO updates without code changes for new pests. Targeted analysis reduces pesticide costs and environmental impact.

6. PROJECT TEAM

Rank	Role in the Team	Education Level	Grade	Member Role
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–	Advisor	Master's Degree	Graduate	Guide
1	Team Leader (Programming Division)	Bachelor's Degree	2nd	Student
2	Member (Mechanical Division)	Bachelor's Degree	2nd	Student
3	Member (Mechanical Division)	Bachelor's Degree	1st	Student
4	Member (Electrical Division)	Bachelor's Degree	2nd	Student
5	Member (Electrical Division)	Bachelor's Degree	1st	Student
6	Member (Non-Technical Research & Development)	Bachelor's Degree	1st	Student

7. SOURCES

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